

LECTURE 11

GROUP AND RING THEORY

8. SMITH NORMAL FORM OVER A PID

Throughout this chapter R is a principal ideal domain with unity. Thus, in particular R may be a field. Recall that the rank of a finitely generated free R -module is defined to be the number of elements in any basis of the free module.

8.1. Row module, column module, and rank. Let $A \in M_{m \times n}(R)$ be an $m \times n$ matrix over R . The rows (respectively columns) of A are the elements of the R -module $R^{1 \times n}$ (respectively $R^{m \times 1}$) consisting of the $1 \times n$ (respectively $m \times 1$) matrices over R . Generally, we write $R^{1 \times n}$ (respectively $R^{m \times 1}$) as R^n (respectively R^m). Using the notation R^n to denote rows as well as columns never creates any confusion because the context always makes the meaning clear.

Definition 8.1. Let $A \in M_{m \times n}(R)$. The submodule of R^n generated by the m rows of A is called the *row module* of A ; and the submodule of R^m generated by the n columns of A is called the *column module* of A .

Let $R(A)$ and $C(A)$, respectively, denote the row module and the column module of the matrix A . If the ring R is a field, the modules $R(A)$ and $C(A)$ are, respectively, called the *row space* and *column space* of the matrix A .

Note that $R(A)$ and $C(A)$ are finitely generated submodules of the free modules R^n and R^m , respectively. Thus by Lemma 5.25, both $R(A)$ and $C(A)$ are free modules.

Definition 8.2. Let $A \in M_{m \times n}(R)$. The rank of the module $R(A)$ (respectively $C(A)$) is called the *row rank* (respectively *column rank*) of A .

The proof of the following theorem is exactly the same as the proof of the corresponding theorem for matrices over a field.

Theorem 8.3. Let $A \in M_{m \times n}(R)$. Let P and Q , respectively, be $m \times m$ and $n \times n$ invertible matrices over R . Then the row (column) rank of PAQ equals the row (column) rank of A .

In order to compute the row rank or the column rank of a given matrix, we reduce the matrix into “normal” form by performing certain operations on rows and columns defined in the next section. Eventually, we obtain by this process that the row rank of a matrix A is equal to the column rank of the matrix.

8.2. Smith normal form. The aforementioned operations are as given below.

Definition 8.4. Let $A \in M_{m \times n}(R)$. The following three types of operations on the rows (respectively columns) of A are called *elementary row* (respectively *column*) *operations*:

- (i) Interchanging two rows (respectively columns). We denote by $R_i \leftrightarrow R_j$ (respectively $C_i \leftrightarrow C_j$) the operation of interchanging the i th and j th rows (respectively columns).

- (ii) Multiplying the elements of one row (respectively column) by a non-zero element of R . We denote by αR_i (respectively αC_i) the operation of multiplying the i th row (respectively column) by α .
- (iii) Adding to the elements of one row (respectively column) α times the corresponding elements of a different row (respectively column), where $\alpha \in R$. We denote by $R_i + \alpha R_j$ (respectively $C_i + \alpha C_j$) the operation of adding to the elements of the i th row (respectively column) α times the corresponding elements of the j th row (respectively column).

The invertible matrices P and Q that we need to consider for Theorem 8.3 are built up of certain matrices, as accounted for below. Recall that the E_{ij} , for $1 \leq i, j \leq n$, denote $n \times n$ matrix units (so with a 1 in position (i, j) and zeros elsewhere), and $E_{ij}E_{kl} = \delta_{jk}E_{il}$, where δ_{jk} is the Kronecker delta. By abuse of notation we also write E_{ij} for the corresponding $m \times m$ matrix, where $1 \leq i, j \leq m$. In the following, it will be clear from the context which E_{ij} is meant.

Theorem 8.5. *Let $A \in M_{m \times n}(R)$.*

- (i) *If $S_{ij} = I - E_{ii} - E_{jj} + E_{ij} + E_{ji}$, then $S_{ij}A$ (respectively AS_{ij}) is the matrix obtained from A by interchanging the i th and the j th rows (respectively columns). Also $S_{ij}^{-1} = S_{ij}$.*
- (ii) *If $M_i(\alpha) = I + (\alpha - 1)E_{ii}$ and α is an invertible element in R , then $M_i(\alpha)A$ (respectively $AM_i(\alpha)$) is the matrix obtained from A by multiplying the i th row (respectively column) by α . Also $M_i^{-1}(\alpha) = I + (\alpha^{-1} - 1)E_{ii}$.*
- (iii) *If $C_{ij}(\alpha) = I + \alpha E_{ij}$ for $\alpha \in R$, then $C_{ij}(\alpha)A$ (respectively $AC_{ji}(\alpha)$) is the matrix obtained from A by multiplying the j th row (respectively column) by α and adding it to the i th row (respectively column). Also $C_{ij}^{-1}(\alpha) = I - \alpha E_{ij}$ for $i \neq j$.*

Proof. (Non-examinable for the oral exam) We only prove the statement for row operations, as the corresponding statements for column operations follow from considering the transpose of the matrices. For example, consider $(AC_{ji}(\alpha))^t = C_{ji}(\alpha)^t A^t = (I + \alpha E_{ji})^t A^t = (I + \alpha E_{ij})^t A^t$ as $E_{ij}^t = E_{ji}$.

(i) $S_{ij}A = (I - E_{ii} - E_{jj} + E_{ij} + E_{ji})A$, or

$$(8.1) \quad S_{ij}A = A - E_{ii}A - E_{jj}A + E_{ij}A + E_{ji}A.$$

We note that

- (a) $E_{ii}A$ is the matrix whose i th row is the same as that of A , and all other rows are zero.
- (b) $E_{ij}A$ is the matrix whose i th row is the j th row of A , and all other rows are zero.

Thus (8.1) shows that $S_{ij}A$ is the matrix obtained from A by interchanging its i th and j th rows. Since $S_{ij}^2 = I$, it follows that $S_{ij}^{-1} = S_{ij}$.

(ii) We have $M_i(\alpha)A = A - E_{ii}A + \alpha E_{ii}A$. Since

$$E_{ii}A = \begin{pmatrix} 0 & 0 & 0 \\ a_{i1} & \cdots & a_{in} \\ 0 & 0 & 0 \end{pmatrix} \leftarrow i\text{th row},$$

we have that $M_i(\alpha)A$ is the matrix obtained from A by multiplying the i th row by α . Further,

$$(I + (\alpha - 1)E_{ii})(I + (\alpha^{-1} - 1)E_{ii}) = I.$$

This proves $M_i^{-1}(\alpha) = (I + (\alpha^{-1} - 1)E_{ii})$.

(iii) Note that $C_{ij}(\alpha)A = A + \alpha E_{ij}A$. As

$$E_{ij}A = \begin{pmatrix} 0 & 0 & 0 \\ a_{j1} & \cdots & a_{jn} \\ 0 & 0 & 0 \end{pmatrix} \leftarrow i\text{th row},$$

we see that $C_{ij}(\alpha)A$ is the matrix obtained from A by multiplying the j th row of A by α and adding it to the i th row. Finally,

$$(I + \alpha E_{ij})(I - \alpha E_{ij}) = I - \alpha E_{ij} + \alpha E_{ij} = I,$$

so $C_{ij}^{-1}(\alpha) = I - \alpha E_{ij}$. $\square$

Definition 8.6. The matrices S_{ij} , $M_i(\alpha)$, and $C_{ij}(\alpha)$ are known as *elementary matrices*.

We note that an elementary matrix is the result of performing a single elementary row or column operation on the identity matrix. Precisely,

S_{ij} = the matrix obtained from the identity matrix by interchanging the i th and j th rows (equivalently, interchanging the i th and j th columns).

$M_i(\alpha)$ = the matrix obtained from the identity matrix by multiplying the i th row by an invertible $\alpha \in R$ (equivalently, by multiplying the i th column by α).

$C_{ij}(\alpha)$ = the matrix obtained from the identity matrix by adding to the elements of the i th row α times the corresponding elements of the j th row, where $\alpha \in R$. (This matrix is also the matrix obtained from the identity matrix by adding to the elements of the j th column α times the corresponding elements of the i th column.)

In addition to these three elementary row (column) operations, we apply a non-elementary operation to the rows and columns of A : multiplication by matrices of the form

$$(iv) \quad \begin{pmatrix} 1 & & & & & \\ & 1 & & & & \\ & & \ddots & & & \\ & & & \begin{pmatrix} u & s \\ v & t \end{pmatrix} & & \\ & & & & 1 & \\ & & & & & 1 & \\ & & & & & & \ddots & \\ & & & & & & & 1 \end{pmatrix},$$

where $\begin{pmatrix} u & s \\ v & t \end{pmatrix}$ is invertible in $M_2(R)$, the 2×2 matrix ring over R .

Multiplying A on the right (respectively left) by a suitable matrix of the above form has the effect of replacing two of the entries on a given row (respectively column) by their greatest common divisor and 0, respectively.

Definition 8.7. Two matrices $A, B \in M_{m \times n}(R)$ are said to be *equivalent* if there exists an invertible matrix $P \in M_m(R)$ and an invertible matrix $Q \in M_n(R)$ such that $B = PAQ$.

It is clear that “being equivalent” defines an equivalence relation in the set of $m \times n$ matrices with entries in R .

We are now ready to state and prove that every matrix is equivalent to a “diagonal” matrix, known as the *Smith normal form*. We will also assume without proof that every element in a principal ideal domain can be factorised into a product of primes (i.e. irreducible elements).

Theorem 8.8. *If $A \in M_{m \times n}(R)$ where R is a principal ideal domain, then A is equivalent to a matrix that has the “diagonal” form*

$$\begin{pmatrix} a_1 & & & & & \\ & a_2 & & & & \\ & & \ddots & & & \\ & & & a_r & & \\ & & & & 0 & \\ & & & & & \ddots \\ & & & & & & 0 \end{pmatrix},$$

where the $a_i \neq 0$ and $a_1 \mid a_2 \mid \cdots \mid a_r$.

Proof. We define the length $\ell(a)$ of $a \neq 0$ to be the number of prime factors occurring in the factorisation $a = p_1 p_2 \cdots p_r$, for p_i primes (not necessarily distinct). We use the convention that $\ell(u) = 0$ if u is a unit.

If $A = 0$, then there is nothing to prove. Otherwise, let a_{ij} be a non-zero element of A with minimal length $\ell(a_{ij})$. Elementary row and column operations bring this element to position $(1, 1)$. We may then assume that a non-zero element of A with smallest length is at position $(1, 1)$.

Suppose first $a_{11} \mid a_{1k}$ for all $2 \leq k \leq n$. In particular $a_{11} \mid a_{12}$. By performing the elementary column operation (iii) on the first two columns (or, equivalently, by multiplying on the right by a suitable elementary matrix), we can reduce A to a matrix whose first row is again of the form $(d \ 0 \ b_{13} \ \cdots \ b_{1n})$, where $\ell(d) \leq \ell(a_{11})$.

Suppose now $a_{11} \nmid a_{1k}$ for some $2 \leq k \leq n$. Interchanging the second and the k th column, we may assume that $a_{11} \nmid a_{12}$. Let $d = (a_{11}, a_{12})$ be the greatest common divisor of a_{11} and a_{12} . Then $\ell(d) < \ell(a_{11})$. There exist elements $u, v \in R$ such that $a_{11}u + a_{12}v = d$. Since $d = (a_{11}, a_{12})$, there exist $s, t \in R$ such that $a_{11} = ds$ and $a_{12} = dt$. Thus $d(us + vt) = d$, and hence $us + vt = 1$. So $\begin{pmatrix} u & t \\ v & -s \end{pmatrix}$ is invertible. Multiplying A on the right by

$$\begin{pmatrix} \begin{pmatrix} u & t \\ v & -s \end{pmatrix} & & & 0 \\ & 1 & & \\ & & 1 & \\ & & & \ddots \\ 0 & & & & 1 \end{pmatrix},$$

we obtain the matrix whose first row is

$$(d \ 0 \ b_{13} \ \cdots \ b_{1n}),$$

where $\ell(d) < \ell(a_{11})$.

Continuing this process yields an equivalent matrix whose first row has all entries 0, except for the entry $(1, 1)$.

Similarly, appropriate elementary row operations (i) - (iii) and the non-elementary operation of multiplying on the left by the matrix of the form given in (iv) reduce the elements

in the first column after position $(1, 1)$ to 0 and either keep the elements in the first row unaltered (i.e. all apart from the $(1, 1)$ entry are zero) or reduce the length of the $(1, 1)$ entry. In the second case we repeat the process by which all the elements in the first row except the one at position $(1, 1)$ are reduced to 0. Since $\ell(a_{11})$ is finite, this process (of alternately reducing the first row and the first column) must come to an end. When it does, we have reduced A to the form

$$P_1 A Q_1 = \begin{pmatrix} a_1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A_1 & \\ 0 & & & \end{pmatrix},$$

where A_1 is an $(m-1) \times (n-1)$ matrix, and P_1 and Q_1 are $m \times m$ and $n \times n$ invertible matrices, respectively.

Similarly, there exist invertible matrices P'_2 and Q'_2 of orders $(m-1) \times (m-1)$ and $(n-1) \times (n-1)$, respectively, such that

$$P'_2 A_1 Q'_2 = \begin{pmatrix} a_2 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & A_2 & \\ 0 & & & \end{pmatrix},$$

where A_2 is an $(m-2) \times (n-2)$ matrix. Let

$$P_2 = \begin{pmatrix} 1 & 0 \\ 0 & P'_2 \end{pmatrix} \quad \text{and} \quad Q_2 = \begin{pmatrix} 1 & 0 \\ 0 & Q'_2 \end{pmatrix}$$

be, respectively, $m \times m$ and $n \times n$ invertible matrices. Then

$$P_2 P_1 A Q_1 Q_2 = \begin{pmatrix} a_1 & 0 & 0 & \cdots & 0 \\ 0 & a_2 & 0 & \cdots & 0 \\ & 0 & & & \\ \vdots & \vdots & & A_2 & \\ 0 & 0 & & & \end{pmatrix}.$$

Continuing like this (or by induction on $m+n$), we obtain

$$PAQ = \text{diag}(a_1, a_2, \dots, a_r, 0, \dots, 0).$$

Finally, we show that we can reduce PAQ further such that $a_1 \mid a_2 \mid \cdots \mid a_r$. Assume $a_1 \nmid a_2$. Add the second row to the first row. The first row then becomes

$$(a_1 \quad a_2 \quad 0 \quad \cdots \quad 0).$$

By performing the operations seen above, we can reduce the length of a_1 . Thus, by further reduction we may assume $a_1 \mid a_2$ and, similarly $a_1 \mid a_i$ for $i = 3, \dots, r$. By repeating this procedure for a_2 in place of a_1 , and so on, we finally reach a situation where $a_i \mid a_{i+1}$, for $i = 1, \dots, r-1$. $\square$

Definition 8.9. The non-zero diagonal elements of the matrix having the diagonal form given in Theorem 8.8 are called the *invariant factors* of A .

It can be shown (beyond the scope of the course) that the invariant factors are unique up to unit multipliers, and two $m \times n$ matrices are equivalent if and only if they have the “same” invariant factors.

Theorem 8.10. *Let A be an $m \times n$ matrix over a PID. Then the row rank of A equals the column rank of A .*

Proof. Choose P and Q invertible matrices of suitable sizes such that PAQ is in Smith normal form. Then by Theorem 8.3,

$$\begin{aligned} \text{row rank}(A) &= \text{row rank}(PAQ) = r = \text{column rank}(PAQ) \\ &= \text{column rank}(A). \end{aligned}$$

□

Definition 8.11. Let A be an $m \times n$ matrix over a PID. The common value of the row rank of A and the column rank of A is called the *rank* of A .

Example 8.12. Obtain the Smith normal form and rank for the matrix with integral entries

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 0 \end{pmatrix}.$$

Solution. We have

$$\begin{aligned} &\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 0 \end{pmatrix} \xrightarrow{R_2 - 4R_1} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -12 \end{pmatrix} \xrightarrow{C_2 - 2C_1, C_3 - 3C_1} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3 & -12 \end{pmatrix} \\ &\xrightarrow{C_3 - 4C_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \end{pmatrix} \xrightarrow{(-1)R_2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \end{pmatrix}. \end{aligned}$$

Clearly $\text{rank}(A) = 2$.

Remark 8.13. Sometimes one may end up with a “diagonal” matrix $\text{diag}(d_1, d_2, \dots)$, where d_i does not necessarily divide d_{i+1} for some i . In such a situation one must continue performing suitable operations until arriving at the desired form.

Example 8.14. Find the invariant factors of the matrix

$$\begin{pmatrix} -x & 4 & -2 \\ -3 & 8-x & 3 \\ 4 & -8 & -2-x \end{pmatrix}$$

over the ring $\mathbb{Q}[x]$. Also find the rank.

Solution. The matrix can be proved to be equivalent to

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & (x-2)(x^2-4x+20) \end{pmatrix}.$$

[We performed (i) $R_1 \leftrightarrow R_3$, (ii) $R_3 + (x/4)R_1$, (iii) $R_2 + \frac{3}{4}R_1$, (iv) $R_3 - 2R_2$, (v) $C_2 + 2C_1$, (vi) $C_3 - \frac{3}{4}C_2$, (vii) $\frac{1}{4}C_1$, (viii) $C_3 + (2+x)C_1$, (ix) $-4C_3$, (x) $R_2 + R_3$, (xi) multiply on the right with

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2-x & x^2-4x+20 \\ 0 & -1 & x-2 \end{pmatrix},$$

(xii) $-\frac{1}{16}R_2$, (xiii) $R_3 + (x^2-4x+20)R_2$]. Hence $\text{rank}(A) = 3$.